SHALEM RAJU

shalemofficial1@gmail.com | Shalem
+91 6282986043 | Shalem-2024

SKILLS

Generative AI & Agents	RAG, LangGraph, Agentic Workflows, Function Calling, Vector Databases (FAISS), State Management, Prompt Engineering.
LLM Engineering	Transformer Architecture, Hugging Face, Tokenization, Context Window Optimization, Semantic Search.
Model Tuning & Ops	Fine-Tuning, PEFT (LoRA/QLoRA), Quantization, Knowledge Distillation, Model Pruning, Adapter Layers, GPU Utilization.
Machine Learning	XGBoost, Scikit-learn, Feature Engineering, Hypothesis Testing, A/B Testing, Model Evaluation .
MLOps & Infrastructure	Docker, Apache Airflow, MLflow, FastAPI, PySpark, Databricks, SQL, AWS EC2, AWS S3

WORK EXPERIENCE

Digital Twin Engineer — Jan 2024 – Present
TVS Motor Company — R&D — Bengaluru, KA

Project: Fine-Tuning Large Language Models for Proprietary Code Generation — Repo

- **Fine-Tuning & Optimization:** Leveraged the **Axolotl** framework to fine-tune **Qwen-2.5-72B** on **H200 GPUs**. Implemented **QLoRA** (Quantized Low-Rank Adaptation) to optimize VRAM usage (∼40GB) while maintaining high-fidelity code generation.
- **Inference Engineering:** Built a production-grade inference script using **vLLM** and **PyTorch**. Engineered a ranked keyword filtering algorithm to dynamically select relevant library objects, solving critical context window bottlenecks.
- **Structured Output Generation:** Solved the challenge of generating strict, proprietary XML schemas from unstructured text by training the model on "Precondition/Action/Postcondition" data pairs.
- **Infrastructure Management:** Deployed and managed ephemeral GPU clusters on **RunPod** (A100/H100), utilizing persistent volume storage for model weight management and reproducible environment setups.

Project: Digital Twin Framework for EV Battery Health Analysis — Repo

- **Architected** a Digital Twin framework on Databricks to analyze battery degradation across **1.6M+ EVs**, transitioning the team from local Jupyter workflows to a scalable **PySpark/ADLS** architecture.
- **Engineered automated ELT pipelines** using **Databricks Jobs** to process 2.5 years of telematics data (219 parameters), reducing data bottlenecks and enabling analysis of 500k+ vehicle cycles.
- **Conducted rigorous Hypothesis Testing** on driver behavior, identifying Depth of Discharge **(DoD)** as the primary driver of SOH degradation, while debunking correlations with speed and braking.
- **Developed an XGBoost regression model** ($MSE: 1.4$) to predict Battery SOH.
- **Influenced Product Strategy** by integrating SOH insights into the iQube mobile app, providing actionable charging recommendations to improve battery longevity and customer satisfaction.

Assistant Manager — Aug 2022 – Jan 2024
Adani Enterprises Ltd — TC — Ahmedabad, GJ

- **Automated** SAP using VBA and Excel Macros to clean and enhance PR data dump, there by reducing the TAT of each buyer by 3 hours.
- **Optimized** procurement processes and vendor selection using insights from previous 15-year datasets there by creating a cost saving of 16 Lakhs.
- Identified data-driven opportunities to improve efficiency and value there by contributing to the action plans set by category leads.

Projects

Full-Stack RAG AI Agent — Repo

- Engineered a stateful AI Agent using **LangGraph** and **FastAPI**, moving beyond simple chat chains to sophisticated, multi-turn conversation flows with persistent memory.
- **Orchestration & ETL:** Implemented a custom Retrieval-Augmented Generation (**RAG**) pipeline using **FAISS** and **OpenAI Embeddings**, enabling the agent to index and query user-uploaded PDFs dynamically.
- **Model Lifecycle Management:** Integrated **OpenAI Function Calling** to create an autonomous system capable of executing external tools based on user intent.
- **Serving:** Orchestrated the entire stack using **Docker Compose** and Nginx reverse proxy, ensuring a production-ready, containerized deployment.

Real-Time Traffic MLOps Pipeline — Repo

- Designed and deployed a containerized machine learning pipeline to predict urban traffic delays, automating the lifecycle from data ingestion to model deployment.
- **Orchestration & ETL:** Developed complex Apache **Airflow** DAGs to ingest real-time data from the TomTom API (5-minute intervals) and perform daily ETL jobs converting raw logs to optimized Parquet files.
- **Model Lifecycle Management:** Integrated **MLflow** for experiment tracking and model registry, implementing an automated "Champion vs. Challenger" promotion logic to ensure only the best-performing models reach production.
- **Serving:** Deployed the production model via a high-performance **FastAPI** microservice, containerized the entire stack using Docker/Docker Compose for consistent environments.

ECU Data Pipeline: DCM to MATLAB — Repo

- **Engineered** an automated ETL tool to convert proprietary automotive data (.DCM) into engineering-ready Excel and MATLAB formats.
- **Developed** a robust RegEx parsing engine to extract complex 2D maps and 1D curves from unstructured text.
- **Built** a zero-config deployment wrapper through VBS/Batch file for portable, one-click execution in Windows environments.
- **Optimized** data integrity by implementing multi-encoding support (UTF-8) and automated comment-stripping.

Education

2018 - 2022	B.Tech in Mechanical Engineering **NIT, Calicut**	(CGPA: 8.03/10)
2015 - 2017	Board of Intermediate at **Aditya jr. college**	(Percentage: 94%)

Last updated: January 18, 2026